# Mixed Traffic Test (Serial)

2026-02-23-12-21-01

## Test Setup Information

| | | |
|---|---|---|
| Overall Setup Info For all Tests | DUT Model | Test_DUT |
| | DUT Firmware | NA |
| | SSID | NETGEAR_2G_wpa2 |
| | Security | wpa2 |
| | No of Devices | 4 (Virtual Clients: 4, Windows: 0, Linux: 0, Mac: 0, Android: 0 ,iOS: 0) |
| | Test Duration (HH:MM:SS) | 00:13:00 |

## Objective

The Candela mixed traffic test is designed to measure the access point performance andstability by running multiple traffic on real clients like Android, Linux, Windows, and IOSconnected to the access point. This test allows the user to choose multiple types of traffic likeclient capacity test, web browser test, video streaming test ping test. Along with theperformance measurements are client connection times, Station 4-Way Handshake time, DHCPtimes, and more. The expected behavior is for the AP to be able to handle all types of traffic onthe several stations (within the limitations of the AP specs) and Make sure all clients can run alltypes of traffic.

## Traffic Details

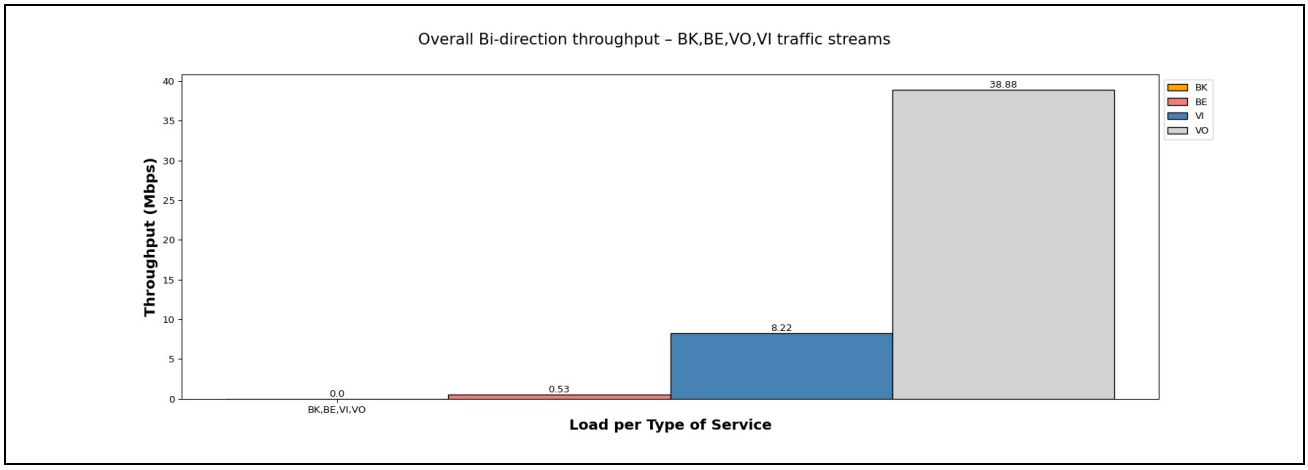
| Sno | Test Cases | Test Duration | Test Status |
|---|---|---|---|
| 1 | Ping Test | | |
| 2 | Quality Of Service(QOS) Test ['VI', 'VO', 'BK', 'BE'] | 1.0 minute | Executed |
| 3 | FTP Test | | |
| 4 | HTTP Test | | |
| 5 | Multicast Test | | |

## 2. Quality Of Service(QOS) Test

## Test Configuration

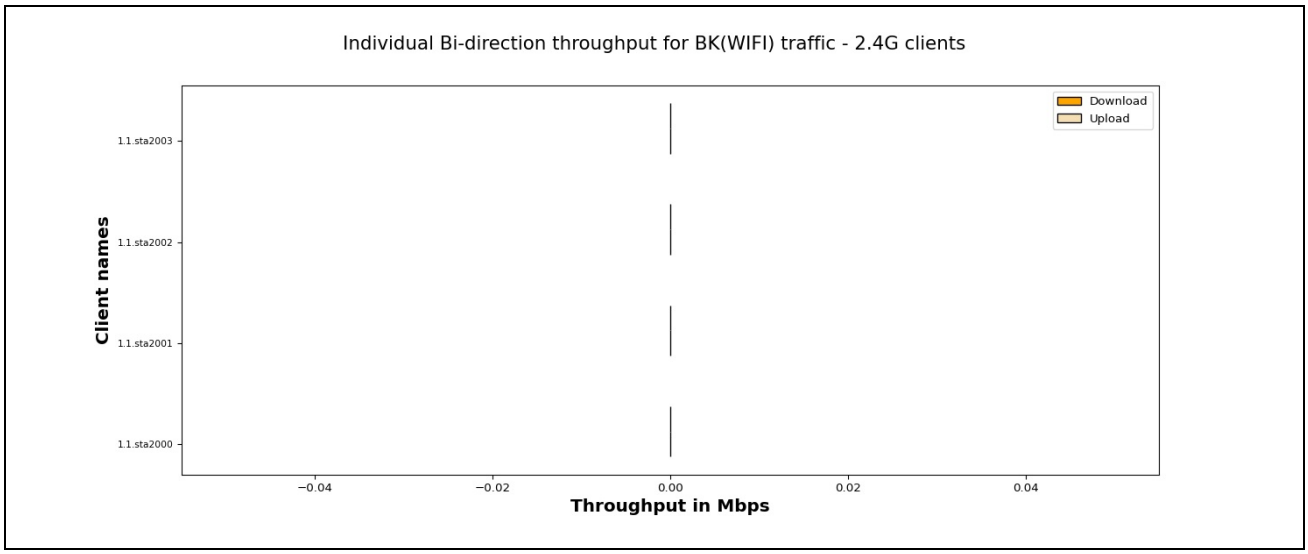
| | | |
|---|---|---|
| Test Setup Information | Protocol | TCP |
| | Traffic Direction | Bi-direction |
| | Security | wpa2 |
| | TOS | ['VI', 'VO', 'BK', 'BE'] |
| | Per TOS Load | Upload:100.0 Mbps,Download:100.0 Mbps |

## Overall Bi-direction Throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)




## Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 4 clients running BK (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

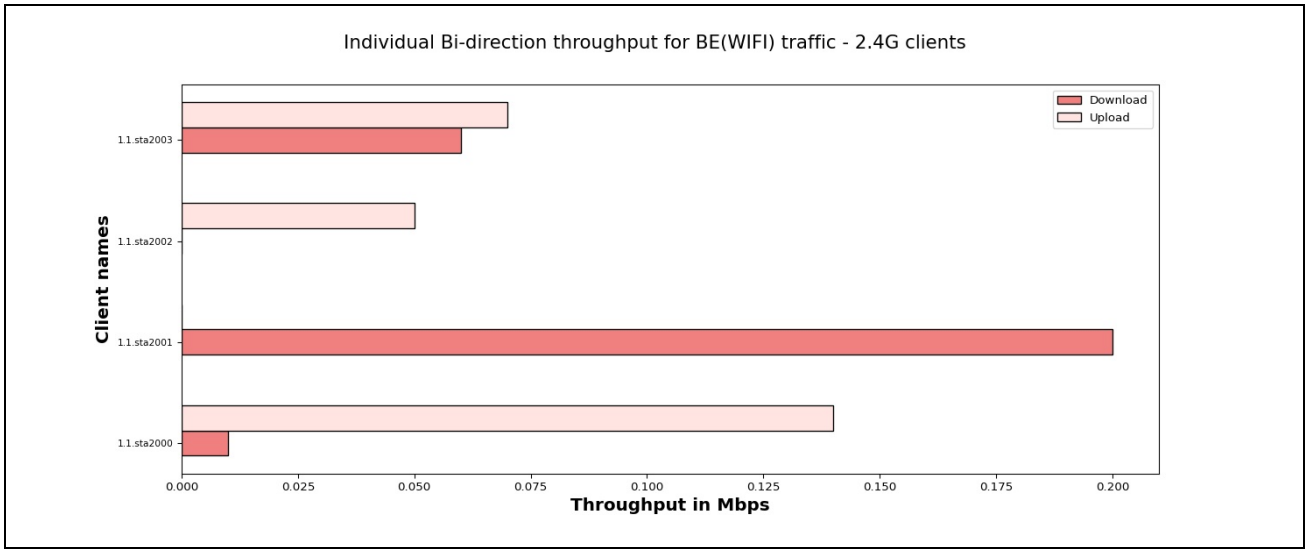



## TOS : Background

| Client Name | Mac | Channel | SSID | Type of traffic | Traffic Direction | Traffic Protocol | Offered upload rate(Mbps) | Offered download rate(Mbps) | Observed upload rate(Mbps) | Observed download rate(Mbps) | Observed Upload Drop (%) | Observed Download Drop (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.sta2000 | 04:f0:21:89:e7:f4 | 11 | NETGEAR_2G_wpa2 | Background | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.0 | 0.0 | 100.0 | 100.0 |
| 1.1.sta2001 | 04:f0:21:89:1a:f4 | 11 | NETGEAR_2G_wpa2 | Background | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.0 | 0.0 | 100.0 | 100.0 |
| 1.1.sta2002 | 04:f0:21:89:20:f4 | 11 | NETGEAR_2G_wpa2 | Background | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.0 | 0.0 | 100.0 | 100.0 |
| 1.1.sta2003 | 04:f0:21:89:7e:f4 | 11 | NETGEAR_2G_wpa2 | Background | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.0 | 0.0 | 75.0 | 100.0 |

## Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 4 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

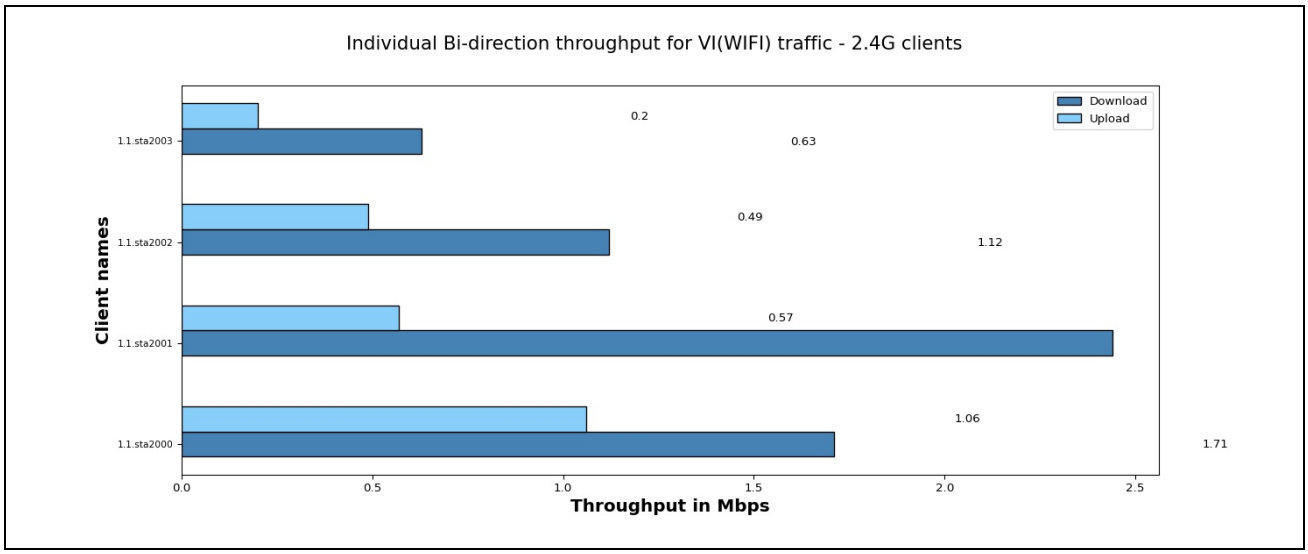



## TOS : Besteffort

| Client Name | Mac | Channel | SSID | Type of traffic | Traffic Direction | Traffic Protocol | Offered upload rate(Mbps) | Offered download rate(Mbps) | Observed upload rate(Mbps) | Observed download rate(Mbps) | Observed Upload Drop (%) | Observed Download Drop (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.sta2000 | 04:f0:21:89:e7:f4 | 11 | NETGEAR_2G_wpa2 | Besteffort | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.14 | 0.01 | 45.454545 | 33.333333 |
| 1.1.sta2001 | 04:f0:21:89:1a:f4 | 11 | NETGEAR_2G_wpa2 | Besteffort | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.00 | 0.20 | 27.272727 | 73.333333 |
| 1.1.sta2002 | 04:f0:21:89:20:f4 | 11 | NETGEAR_2G_wpa2 | Besteffort | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.05 | 0.00 | 41.666667 | 23.529412 |
| 1.1.sta2003 | 04:f0:21:89:7e:f4 | 11 | NETGEAR_2G_wpa2 | Besteffort | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.07 | 0.06 | 57.142857 | 50.000000 |

## Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 4 clients running VI (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

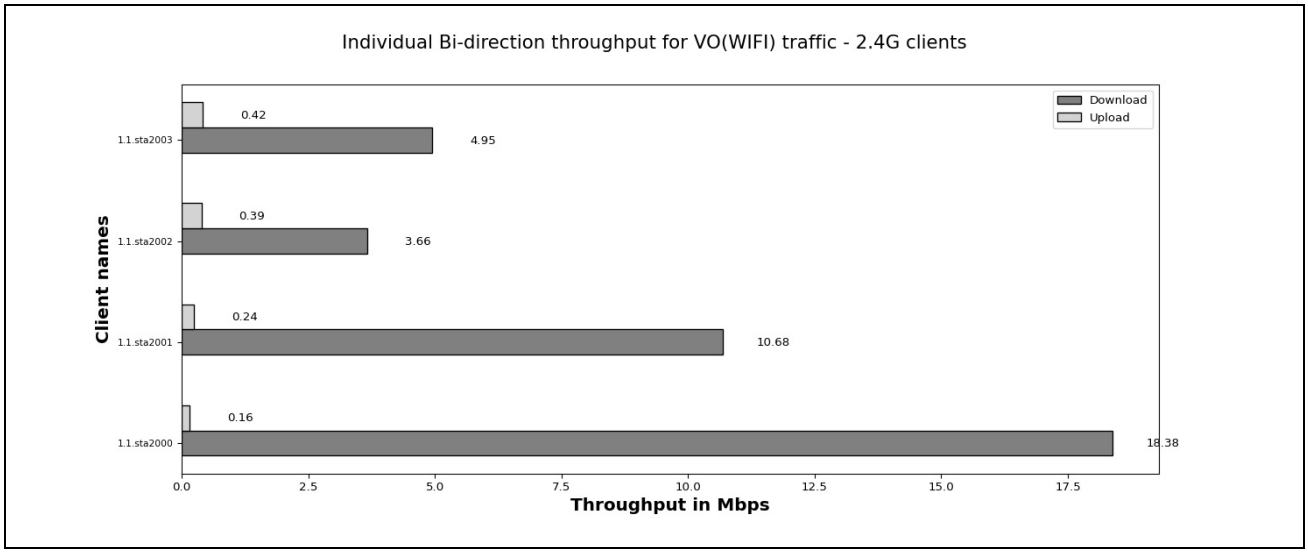



## TOS : Video

| Client Name | Mac | Channel | SSID | Type of traffic | Traffic Direction | Traffic Protocol | Offered upload rate(Mbps) | Offered download rate(Mbps) | Observed upload rate(Mbps) | Observed download rate(Mbps) | Observed Upload Drop (%) | Observed Download Drop (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.sta2000 | 04:f0:21:89:e7:f4 | 11 | NETGEAR_2G_wpa2 | Video | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 1.06 | 1.71 | 29.319372 | 25.423729 |
| 1.1.sta2001 | 04:f0:21:89:1a:f4 | 11 | NETGEAR_2G_wpa2 | Video | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.57 | 2.44 | 14.563107 | 11.872146 |
| 1.1.sta2002 | 04:f0:21:89:20:f4 | 11 | NETGEAR_2G_wpa2 | Video | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.49 | 1.12 | 25.252525 | 21.965318 |
| 1.1.sta2003 | 04:f0:21:89:7e:f4 | 11 | NETGEAR_2G_wpa2 | Video | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.20 | 0.63 | 15.789474 | 24.050633 |

## Individual Bi-direction throughput with intended load Upload:100.0 Mbps,Download:100.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 4 clients running VO (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".




## TOS : Voice

| Client Name | Mac | Channel | SSID | Type of traffic | Traffic Direction | Traffic Protocol | Offered upload rate(Mbps) | Offered download rate(Mbps) | Observed upload rate(Mbps) | Observed download rate(Mbps) | Observed Upload Drop (%) | Observed Download Drop (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.sta2000 | 04:f0:21:89:e7:f4 | 11 | NETGEAR_2G_wpa2 | Voice | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.16 | 18.38 | 26.470588 | 13.766389 |
| 1.1.sta2001 | 04:f0:21:89:1a:f4 | 11 | NETGEAR_2G_wpa2 | Voice | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.24 | 10.68 | 21.153846 | 1.702128 |
| 1.1.sta2002 | 04:f0:21:89:20:f4 | 11 | NETGEAR_2G_wpa2 | Voice | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.39 | 3.66 | 15.151515 | 30.793651 |

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.sta2003 | 04:f0:21:89:7e:f4 | 11 | NETGEAR_2G_wpa2 | Voice | Bi-direction | TCP | 100.0 Mbps | 100.0 Mbps | 0.42 | 4.95 | 9.523810 | 10.110803 |

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|---|---|---|
| Overall Info | contact | support@candelatech.com |